

## Lab 03 Text File I/O

### Objective

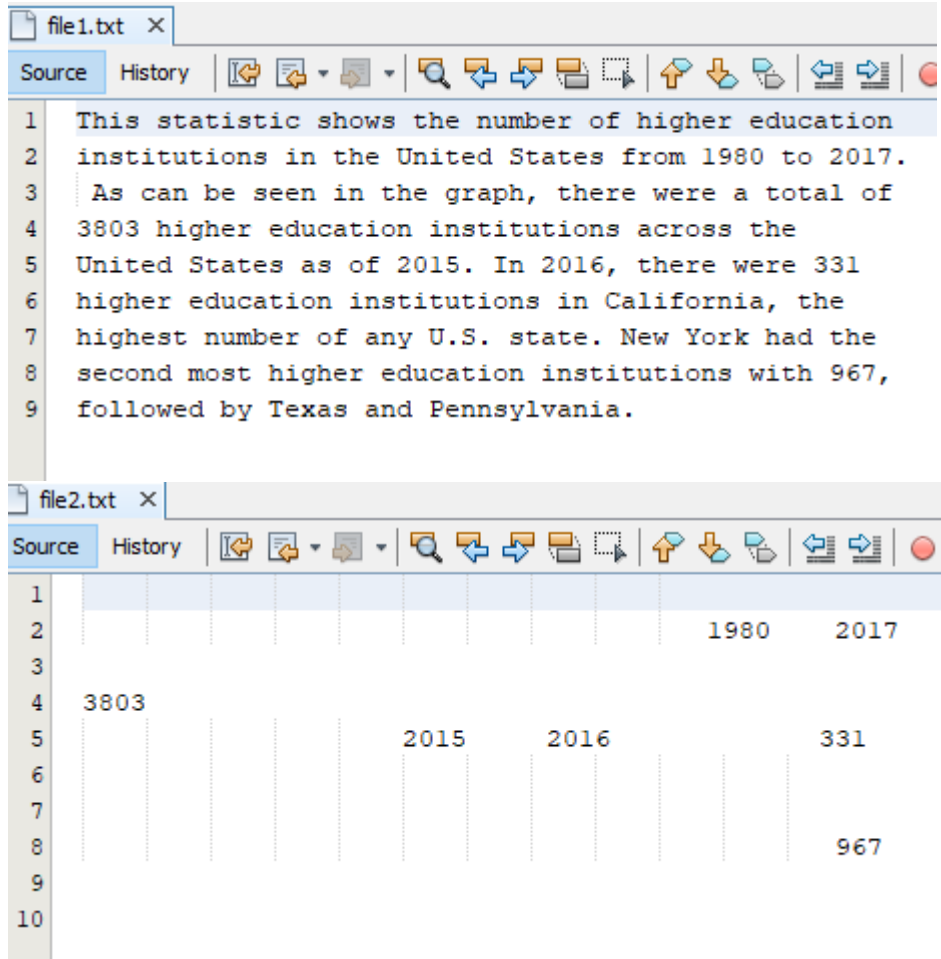
In this lab, you will learn how to:

- File Processing using the File class
- Discovering file/directory properties, to using the File class
- Read /write data from/to a file using the Scanner/PrintWriter class.

### Activity Description

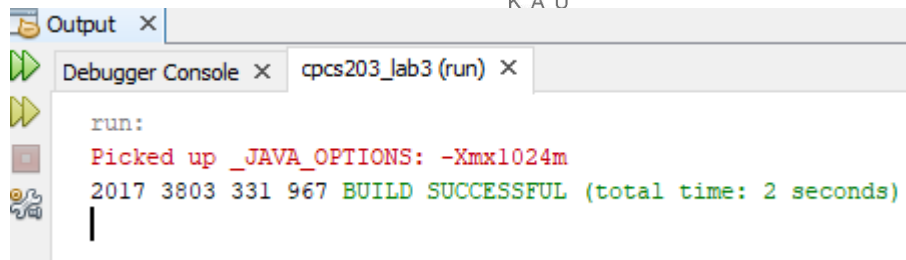
Create a java application called [FileIOExample](#). The application should read contents from first file and extract only numbers from it and write those numbers to the second file. Then the application should then read all numbers from the second file and display only the prime numbers to the terminal.

- There are two files used in this program, first file “[file1.txt](#)” containing text and digits and another file “[file2.txt](#)” to write only the extracted numbers from the first file.



```
file1.txt x
Source History | [Icons]
1 This statistic shows the number of higher education
2 institutions in the United States from 1980 to 2017.
3 As can be seen in the graph, there were a total of
4 3803 higher education institutions across the
5 United States as of 2015. In 2016, there were 331
6 higher education institutions in California, the
7 highest number of any U.S. state. New York had the
8 second most higher education institutions with 967,
9 followed by Texas and Pennsylvania.

file2.txt x
Source History | [Icons]
1
2 1980 2017
3
4 3803
5 2015 2016 331
6
7
8 967
9
10
```



- When finished, make sure that the data written to the new file has been flushed from its buffer and that any system resources used during for running your code have been released.
- **Note** Check the input file existence before start reading from the input files, if the file does not exist, display a message then exit the program execution.

### Hints:

- The **File** and **PrintWriter** classes: exist in (java.io. \*) package.
- You need to open input/output files for reading and writing using:

```
File inputFile = new File("file1.txt");  
File outputFile = new File("file2.txt");  
File inputFile1 = new File("file2.txt");
```

- Make **Scanner** variable to read from input file and make **Printwriter** variable to write to output file using:

```
Scanner input = new Scanner(inputFile);  
PrintWriter output = new PrintWriter(outputFile);  
Scanner input1 = new Scanner(inputFile1);
```

Use **close()** method after completing the process of reading and writing operations on the file.